// ==

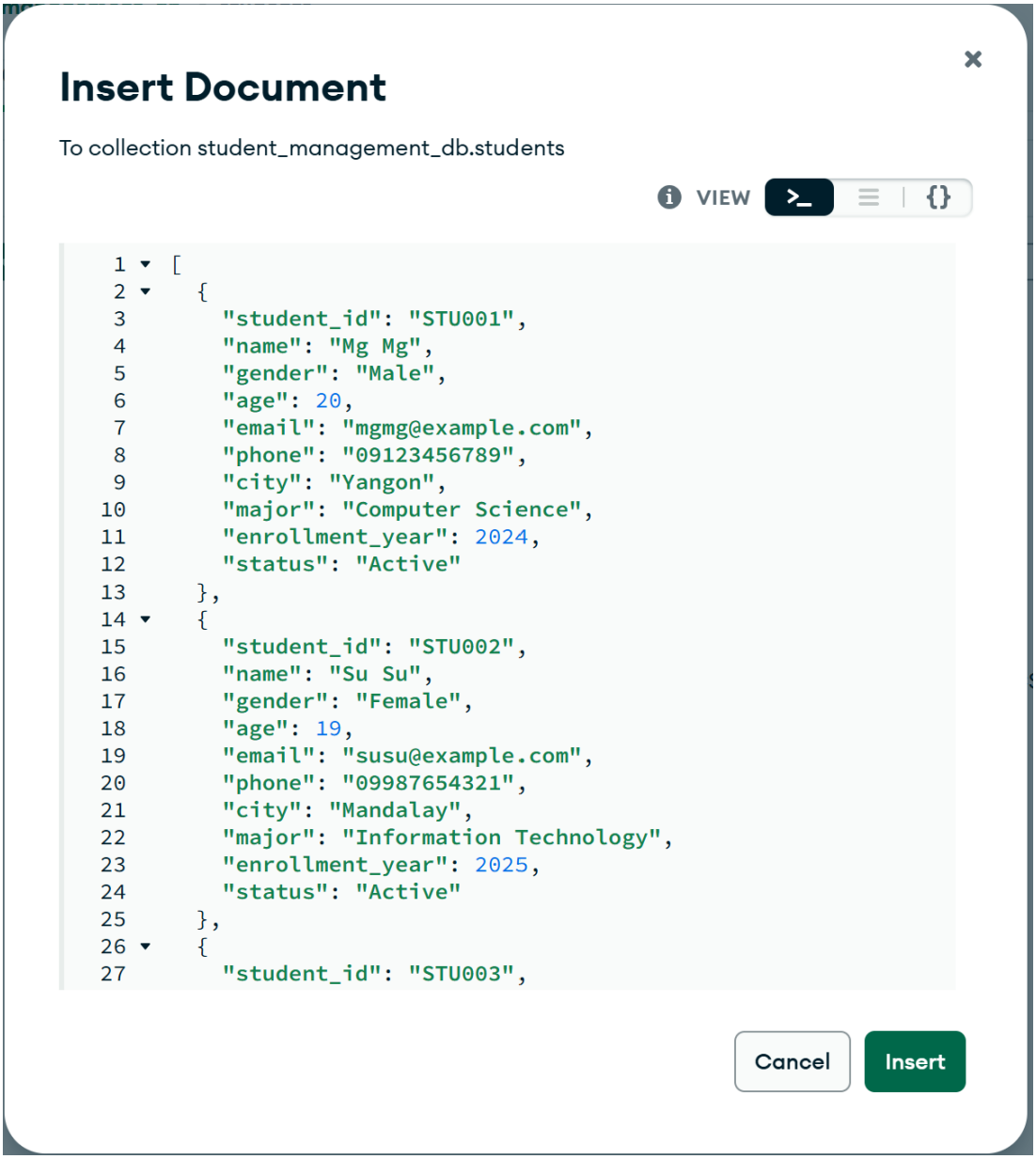
// Student Management System - MongoDB Lab

// ==

// 1. Create / use database

```
> use("student_management_db");
< switched to db student_management_db
```

// 2. Create students collection and insert data



// 3. Create courses collection

```
> db.courses.insertMany([
    {
      course_id: "C001",
      course_name: "Database Systems",
      credit: 3,
      instructor: "Dr. Hla Hla"
    },
    {
      course_id: "C002",
      course_name: "Web Development",
      credit: 3,
      instructor: "Dr. Kyaw Kyaw"
    },
    {
      course_id: "C003",
      course_name: "Data Structures",
      credit: 4,
      instructor: "Dr. Myint Myint"
    }
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace17da4886783d50a832c6'),
      '1': ObjectId('6ace17da4886783d50a832c7'),
      '2': ObjectId('6ace17da4886783d50a832c8')
    }
  }
```

// 4. Create enrollments collection

>_MONGOSH

```
> db.enrollments.insertMany([
    {
      enrollment_id: "E001",
      student_id: "STU001",
      course_id: "C001",
      semester: "Semester 1",
      academic_year: "2025-2026",
      marks: 85,
      grade: "A"
    },
    {
      enrollment_id: "E002",
      student_id: "STU001",
      course_id: "C002",
      semester: "Semester 1",
      academic_year: "2025-2026",
      marks: 78,
      grade: "B"
    },
    {
      enrollment_id: "E003",
      student_id: "STU002",
      course_id: "C001",
      semester: "Semester 1",
      academic_year: "2025-2026",
      marks: 92,
      grade: "A"
    },
    {
      enrollment_id: "E004",
      student_id: "STU003",
      course_id: "C003",
      semester: "Semester 2",
      academic_year: "2025-2026",
      marks: 65,
      grade: "C"
    }
]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace17fd4886783d50a832c9'),
      '1': ObjectId('6ace17fd4886783d50a832ca'),
      '2': ObjectId('6ace17fd4886783d50a832cb'),
      '3': ObjectId('6ace17fd4886783d50a832cc')
    }
}
```

// ==

// Basic Query Lab

// ==

// 5. Show all students

```
>_MONGOSH
> db.students.find();
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09123456789',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ed'),
    student_id: 'STU002',
    name: 'Su Su',
    gender: 'Female',
    age: 19,
    email: 'susu@example.com',
    phone: '09987654321',
    city: 'Mandalay',
    major: 'Information Technology',
    enrollment_year: 2025,
    status: 'Active'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ee'),
    student_id: 'STU003',
    name: 'Aung Aung',
    gender: 'Male',
    age: 21,
    email: 'aungaung@example.com',
    phone: '09777777777',
    city: 'Nay Pyi Taw',
    major: 'Software Engineering',
    enrollment_year: 2023,
    status: 'Inactive'
  }
```

// 6. Show students in Yangon

```
> db.students.find({ city: "Yangon" });
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09123456789',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active'
  }
```

// 7. Show active students only

```
> db.students.find({ status: "Active" });
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09123456789',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ed'),
    student_id: 'STU002',
    name: 'Su Su',
    gender: 'Female',
    age: 19,
    email: 'susu@example.com',
    phone: '09987654321',
    city: 'Mandalay',
    major: 'Information Technology',
    enrollment_year: 2025,
    status: 'Active'
  }
```

// 8. Find one student by student_id

```
> db.students.findOne({ student_id: "STU001" });
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09123456789',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active'
  }
```

// 9. Show only name, major, city

```
> db.students.find(
    {},
    { _id: 0, name: 1, major: 1, city: 1 }
  );
< {
    name: 'Mg Mg',
    city: 'Yangon',
    major: 'Computer Science'
  }
  {
    name: 'Su Su',
    city: 'Mandalay',
    major: 'Information Technology'
  }
  {
    name: 'Aung Aung',
    city: 'Nay Pyi Taw',
    major: 'Software Engineering'
  }
```

// 10. Students older than 19

```
> db.students.find({ age: { $gt: 19 } });
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09123456789',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ee'),
    student_id: 'STU003',
    name: 'Aung Aung',
    gender: 'Male',
    age: 21,
    email: 'aungaung@example.com',
    phone: '09777777777',
    city: 'Nay Pyi Taw',
    major: 'Software Engineering',
    enrollment_year: 2023,
    status: 'Inactive'
  }
```

// ==

// Update Lab

// ==

// 11. Update student phone number

```
> db.students.updateOne(
    { student_id: "STU001" },
    { $set: { phone: "09999999999" } }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

// 12. Change inactive student to active

```
> db.students.updateOne(
    { student_id: "STU003" },
    { $set: { status: "Active" } }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

// 13. Add address field

```
> db.students.updateMany(
    {},
    { $set: { country: "Myanmar" } }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
```

// ==========

// Delete Lab

// ==========

// 14. Delete one inactive student

```
> db.students.deleteOne({ student_id: "STU003" });
< {
    acknowledged: true,
    deletedCount: 1
  }
```

// ==

// Sorting and Limiting Lab

// ==

// 15. Sort students by age ascending

```
> db.students.find().sort({ age: 1 });
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ed'),
    student_id: 'STU002',
    name: 'Su Su',
    gender: 'Female',
    age: 19,
    email: 'susu@example.com',
    phone: '09987654321',
    city: 'Mandalay',
    major: 'Information Technology',
    enrollment_year: 2025,
    status: 'Active',
    country: 'Myanmar'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09999999999',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active',
    country: 'Myanmar'
  }
```

// 16. Sort students by enrollment year descending

```
> db.students.find().sort({ enrollment_year: -1 });
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ed'),
    student_id: 'STU002',
    name: 'Su Su',
    gender: 'Female',
    age: 19,
    email: 'susu@example.com',
    phone: '09987654321',
    city: 'Mandalay',
    major: 'Information Technology',
    enrollment_year: 2025,
    status: 'Active',
    country: 'Myanmar'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09999999999',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active',
    country: 'Myanmar'
  }
```

// 17. Show first 2 students only

```
> db.students.find().limit(2);
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09999999999',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active',
    country: 'Myanmar'
  }
  {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ed'),
    student_id: 'STU002',
    name: 'Su Su',
    gender: 'Female',
    age: 19,
    email: 'susu@example.com',
    phone: '09987654321',
    city: 'Mandalay',
    major: 'Information Technology',
    enrollment_year: 2025,
    status: 'Active',
    country: 'Myanmar'
  }
```

// ==

// Aggregation Lab

// ==

// 18. Count students by city

```
> db.students.aggregate([
    {
      $group: {
        _id: "$city",
        total_students: { $sum: 1 }
      }
    }
  ]);
< {
    _id: 'Mandalay',
    total_students: 1
  }
  {
    _id: 'Yangon',
    total_students: 1
  }
```

// 19. Average marks by course

```
> db.enrollments.aggregate([
    {
      $group: {
        _id: "$course_id",
        average_marks: { $avg: "$marks" }
      }
    }
  ]);
< {
    _id: 'C003',
    average_marks: 65
  }
  {
    _id: 'C001',
    average_marks: 88.5
  }
  {
    _id: 'C002',
    average_marks: 78
  }
```

// 20. Students with marks greater than 80

```
> db.enrollments.find({ marks: { $gt: 80 } });
< {
    _id: ObjectId('6ace17fd4886783d50a832c9'),
    enrollment_id: 'E001',
    student_id: 'STU001',
    course_id: 'C001',
    semester: 'Semester 1',
    academic_year: '2025-2026',
    marks: 85,
    grade: 'A'
  }
  {
    _id: ObjectId('6ace17fd4886783d50a832cb'),
    enrollment_id: 'E003',
    student_id: 'STU002',
    course_id: 'C001',
    semester: 'Semester 1',
    academic_year: '2025-2026',
    marks: 92,
    grade: 'A'
  }
```

// ==

// Join Lab using $lookup

// ==

// 21. Join students with enrollments

>_MONGOSH

```
> db.students.aggregate([
    {
      $lookup: {
        from: "enrollments",
        localField: "student_id",
        foreignField: "student_id",
        as: "enrollment_details"
      }
    }
  ]);
< {
    _id: ObjectId('6ace17c9fbf6d8296f92c6ec'),
    student_id: 'STU001',
    name: 'Mg Mg',
    gender: 'Male',
    age: 20,
    email: 'mgmg@example.com',
    phone: '09999999999',
    city: 'Yangon',
    major: 'Computer Science',
    enrollment_year: 2024,
    status: 'Active',
    country: 'Myanmar',
    enrollment_details: [
      {
        _id: ObjectId('6ace17fd4886783d50a832c9'),
        enrollment_id: 'E001',
        student_id: 'STU001',
        course_id: 'C001',
```

// 22. Join enrollments with courses

```
> db.enrollments.aggregate([
    {
      $lookup: {
        from: "courses",
        localField: "course_id",
        foreignField: "course_id",
        as: "course_details"
      }
    }
  ]);
< {
    _id: ObjectId('6ace17fd4886783d50a832c9'),
    enrollment_id: 'E001',
    student_id: 'STU001',
    course_id: 'C001',
    semester: 'Semester 1',
    academic_year: '2025-2026',
    marks: 85,
    grade: 'A',
    course_details: [
      {
        _id: ObjectId('6ace17da4886783d50a832c6'),
        course_id: 'C001',
        course_name: 'Database Systems',
        credit: 3,
        instructor: 'Dr. Hla Hla'
      }
    ]
  }
```

// ==

// Index Lab

// ==

// 23. Create index on student_id

```
> db.students.createIndex({ student_id: 1 });
< student_id_1
```

// 24. Create unique index on email

```
> db.students.createIndex({ email: 1 }, { unique: true });
< email_1
```

// 25. View indexes

```
> db.students.getIndexes();
< [
    { v: 2, key: { _id: 1 }, name: '_id_' },
    { v: 2, key: { student_id: 1 }, name: 'student_id_1' },
    { v: 2, key: { email: 1 }, name: 'email_1', unique: true }
  ]
```

// ==

// Cleanup Lab

// ==

// 26. Drop one collection

```
> db.enrollments.drop();
< true
```

// 27. Drop database

```
> db.dropDatabase();
< { ok: 1, dropped: 'student_management_db' }
```